Elangoraj Thiruppandiaraj

Senior Data Scientist – AI | GenAI, RAG & Insurance Analytics | AI Consulting

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Professional Summary

Senior Data Scientist – AI with 10 years of experience designing, prototyping, and scaling AI, ML, and Generative AI solutions across insurance, financial services, HR technology, and enterprise environments. Strong hands-on expertise in RAG, agentic workflows, prompt engineering, document understanding, vector search, model evaluation, Python, SQL, Spark, MLOps, and cloud deployment. Experienced in translating ambiguous business problems into working prototypes, production-ready applications, and reusable AI-enabled workflows. Proven ability to work in consulting-style engagements, lead discovery workshops, mentor engineers, and partner with senior stakeholders across underwriting, risk, compliance, product, and technology teams.

Key Skills & Technical Stack

- **GenAI & AI Application Development:** RAG, Prompt Engineering, Tool-Augmented Agents, Agentic Workflows, Multi-Agent Systems, Document Understanding, Natural Language Interfaces, Structured Output Generation, LLM Evaluation.
- **Data Science & Machine Learning:** Predictive Modelling, NLP, OCR, Recommendation Systems, Anomaly Detection, Time Series, Supervised/Unsupervised Learning, Model Evaluation, A/B Testing, Statistical Analysis.
- **AI Engineering & MLOps:** Python, SQL, Spark, MLflow, Docker, GitHub, CI/CD, Model Deployment, Monitoring, Vector Databases, Embeddings, Streamlit, Databricks.
- **Insurance & Risk Domain:** Insurance, Underwriting, Risk Assessment, Financial Services, Compliance, Data Governance, Responsible AI, Privacy-by-Design.
- **Consulting & Leadership:** Technical Discovery, AI Capability Reviews, AI Roadmaps, Client/Stakeholder Advisory, Proposal Support, Workstream Ownership, Mentoring, Executive Communication.
- **Cloud Platforms:** Azure ML, Azure AI Foundry, Azure Databricks, Azure App Services, Azure DevOps, GCP Vertex AI, BigQuery, AWS S3, EMR, SageMaker, Redshift.

Work History

Data Scientist (AI Solution Lead) | Just Group Plc, London, UK | July 2024- Present

- Lead the design, prototyping, and delivery of applied AI and GenAI solutions across underwriting, risk, and retail, translating ambiguous business problems into working prototypes, reusable AI workflows, and production-ready applications. Manage and mentor a team of 3 AI/ML engineers, providing technical leadership through architecture reviews, evaluation standards, and delivery oversight.
- **Document Intelligence & RAG Platform:** Designed and deployed a secure document understanding system used by medical underwriters to analyse complex reports and financial documents, automating highly-structured decision-making workflows similar to **credit and risk rating processes**. Built a multimodal pipeline combining OCR-based extraction from scanned images and handwritten notes, vector embeddings for semantic indexing, similarity search across document chunks, and re-ranking strategies to retrieve high-confidence answers. The platform processes structured and unstructured medical and financial data and is being expanded to support additional business workflows.
- **Agentic AI Platform:** Architected and delivered a multi-agent Generative AI system enabling asynchronous orchestration of multiple GPT-family models to generate structured outputs and automatically populate template-based documents. The system processes transcripts, PDFs, and

structured inputs to produce refined reports and operational notes, saving hundreds of hours of manual effort each month across multiple business teams.

- Led discovery workshops and consulting-style engagements with underwriting, risk, compliance, and business stakeholders to identify AI opportunities, assess feasibility, define roadmaps, and translate domain expertise into machine-readable rules, checks, and evaluation criteria.
- Deployed the platform using Azure App Services with CI/CD pipelines in Azure DevOps, enabling scalable internal access and automated model updates.
- **Anomaly Detection for Financial Data:** Developed a similarity-search–based anomaly detection approach using embedding representations to identify irregular patterns within quote and underwriting datasets. The system detects subtle behavioural deviations and data inconsistencies that traditional rule-based checks fail to capture, improving early detection of potential risk or fraud-like patterns.
- Partner with business, IT, and compliance teams to ensure AI solutions align with regulatory and governance frameworks; regularly present AI architecture decisions, roadmap progress, and ROI metrics to the **Group Executive Committee** to secure continued investment in AI capabilities.
- Founded and lead the **AI4U initiative and AI Centre of Excellence activities**, driving AI adoption through use-case discovery, prompt engineering workshops, capability sessions, and internal consulting, reaching **200+ active participants and over 40% of the organisation including executive leadership**.
- Represent Just Group externally as an AI speaker at industry conferences including **ai:NOW 2025 and IRM Europe 2025**, presenting on Responsible AI, AI in Financial Services, and Agentic AI systems.

Product Manager | Mercer, London, UK | Dec 2021- June 2024

- Owned the roadmap and end-to-end delivery of an AI-driven internal product, operating at the intersection of product, data science, and engineering. Led discovery workshops with business stakeholders to define requirements and translate them into scalable ML solutions.
- Managed and mentored a cross-functional team of two AI/Data Engineers, acting as the bridge between business stakeholders and technical teams to ensure alignment between product strategy, technical feasibility, and delivery outcomes.
- Designed and implemented a multilingual transformer-based semantic matching system using encoder-based architectures and contextual embeddings from Hugging Face models, improving matching accuracy by **73%** and reducing operational costs by **33%**.
- Led experimentation and optimisation of transformer models using **Parameter Efficient Fine-Tuning (PEFT)**, achieving a **15% performance improvement** over base models.
- Established **MLOps practices** including model tracking, versioning, and reproducibility using MLflow and cloud infrastructure, while exploring Generative AI approaches such as **RAG, RLHF, and LangChain** to improve contextual intelligence across products.

Research Intern (AI) | Intelligent AI, Exeter, UK | April – May 2021

- Developed a satellite image scene-classification model to detect ten land-use categories (e.g., rivers, residences, forests) using Python, PyTorch/TensorFlow, and Docker.

Module lead | LTIMindtree, Chennai, India | Jan 2016 - Oct 2021

- Led the development of a deep neural network–based recommendation engine for Unilever using Apache Spark and Python, supported by early MLOps practices for experiment tracking, model evaluation, and deployment workflows. Improving product placement accuracy to **80%** and ensuring optimal SKU distribution across retail outlets.
- Built and maintained **data engineering pipelines using SQL, Python, and Apache Spark**, processing large retail datasets for machine learning training, analytics, and operational reporting.
- Developed **Power BI dashboards and reporting solutions** for business stakeholders, translating complex data insights into actionable recommendations for supply chain and sales teams.

- Delivered **internal training sessions on machine learning and data science**, contributing to wider adoption of ML/DL practices across engineering teams as part of the Data Science Centre of Excellence.
- Worked onsite at client locations for over **one year**, collaborating directly with business stakeholders to gather requirements, perform data processing, and build reporting and analytics solutions aligned with operational needs.

Publication

Remote Sensing Single Image Super-Resolution Benchmarking with Transfer Learning Algorithms, 2023, Publisher: IEEE, Co-Author: Saptarshi Das.

Education

MSc Data Science with Artificial Intelligence (Distinction) University of Exeter, UK | 2020-2021
Bachelors in Electronics and Communication St. Joseph's College of Engineering, India | 2011-2015

Certifications

- Google Project Management, Generative AI with Large Language Model
- Technical Team Leadership (**University of Oxford),** Deep learning Nanodegree

LEADERSHIP & COMMUNITY

- **Vice President Education, Toastmasters International (City of London)** – Lead club education, meetings, support member development, and drive engagement. Awarded the Future Forward Award for advancing member growth and club performance.
- **AI Mentor & Career Coach** – Support professionals transitioning into AI and tech through mentoring, external coaching, and Fiverr, focusing on learning paths, portfolios, and job.

Languages: Tamil, English